

Comprehensive Project Report: Electric Vehicle (EV) Market & Performance Analysis

Introduction

The automotive industry is shifting from internal combustion engine (ICE) vehicles to sustainable electric mobility. Driven by global climate goals, stricter emission standards, and rapid advances in battery and powertrain engineering, Electric Vehicles (EVs) have become central to this transition.

Understanding today's EV landscape requires evaluating key technical specs, including battery capacity, driving range, efficiency, top speed, and fast-charging capability. This project analyzes a dataset of modern electric vehicles (up to model year 2025) using data analysis and visualization to identify engineering trade-offs, market segments, and performance benchmarks.

Abstract

- **Scope & Dataset Overview:** Exhaustive exploration of the 2025 global electric vehicle market using a dataset of 478 vehicle specifications across numerous international brands. Evaluates key metrics including driving range, battery capacity, energy efficiency, top speed, and DC fast-charging power through rigorous data cleaning, missing value handling, and exploratory data analysis (EDA).
- **Top Performers & Benchmarks:**
 - **Driving Range:** Mercedes-Benz (EQS 450+ 685 km) and Lucid (Air Grand Touring 665 km) achieve the highest overall driving ranges.
 - **Battery Capacity:** Lotus, Rolls-Royce, Cadillac, and Voyah lead among top brands providing superior battery capacity.
 - **Energy Efficiency:** Dacia Spring Electric 45, Fiat 500e Hatchback 24 kWh, Fiat 500e 3+1 24 kWh, and Fiat 500e Cabrio 24 kWh lead in energy efficiency.
 - **Fast Charging:** Top fast-charging capabilities are spearheaded by brands like Porsche and Lotus.
 - **Top Speed:** Maserati GranTurismo Folgore (325 km/h) and Porsche Taycan Turbo GT Weissach (305 km/h) stand out as top vehicles for speed.
- **Engineering & Technical Correlations:**
 - **Battery vs. Range:** Strong positive correlation (0.88043) between battery capacity and driving range, establishing energy storage as the primary driver for extended distance capabilities.
 - **Battery vs. Fast Charging:** Robust linear relationship exists between battery size and fast-charging power, indicating larger packs support higher peak charging rates.
 - **Efficiency Decoupling:** Energy efficiency remains largely decoupled from raw driving range, allowing compact urban EVs and luxury performance vehicles to occupy distinct efficiency tiers.
- **Drivetrain Performance Comparison:**

- **AWD:** Leads in average driving range (~450 km), top speed (~210 km/h), and acceleration (~4.6s).
- **RWD:** Averages ~420 km driving range, ~185 km/h top speed, and ~6.9s acceleration.
- **FWD:** Averages ~300 km driving range, ~155 km/h top speed, and ~9.7s acceleration.
- **Charging Standards & Infrastructure:** CCS dominates as the industry standard accounting for 99.8% of vehicles, while CHAdeMO accounts for only 0.2%, reflecting its gradual phase-out.
- **Cargo Capacity & Body Types:** Small Passenger Vans offer the largest average capacity (~899 liters), whereas Hatchbacks and Cabriolets provide the least, with Cabriolets lowest at ~203 liters.

Market Segments & Conclusion: SUVs dominate overall market volume, while luxury and executive segments lead in average driving range. Synthesized insights serve as a robust benchmarking framework for researchers, manufacturers, and consumers.

Objectives

The primary objectives of this project are:

- **Market Benchmarking:** To evaluate and rank global EV manufacturers and models based on key performance indicators (KPIs) including driving range, battery capacity, energy efficiency, and top speed.
- **Technical Relationship Analysis:** To investigate core engineering correlations, specifically how battery capacity scales with driving range and fast-charging power.
- **Segment & Body-Type Distribution:** To identify dominant vehicle body types, drivetrain configurations, and market segment preferences across the global EV ecosystem.
- **Data-Driven Reporting:** To deliver a structured, reproducible analytical pipeline and report outlining actionable insights into modern EV capabilities.

Methodology

The project followed a structured data science workflow, moving from raw data ingestion to advanced visualization and reporting:

1. **Data Collection & Inspection:**
 - Ingested a multi-attribute CSV dataset comprising 478 entries and 22 technical columns detailing vehicle specifications.
 - Inspected data types, summary statistics, and initial distributions to understand feature scales and value ranges.
2. **Data Cleaning & Imputation:**
 - Handled missing values systematically: imputed categorical/port entries (e.g., standardizing missing charging ports based on brand mode) and applied median-based group imputations where statistically viable.
 - Coerced unstructured fields (such as cargo volume) into numeric data types to facilitate quantitative evaluation.
3. **Exploratory Data Analysis (EDA) & Statistical Correlation:**

- Computed correlation matrices to quantify relationships between continuous variables (e.g., battery size vs. range, charging power vs. battery capacity).
 - Grouped and aggregated data by brand, vehicle segment, and body type to extract top-performing models and market trends.
4. **Data Visualization:**
- Generated distribution plots, correlation heatmaps, joint plots with hexagonal binning, and regression trend lines (with confidence intervals) using Python data visualization libraries to clearly interpret multi-variable dynamics.

Tools Used

The execution of this project relied on an integrated stack of data analysis, visualization, and business intelligence tools:

- **Programming Language:** Python
- **Data Manipulation & Numerical Computing:** pandas, numpy
- **Data Visualization:** matplotlib, seaborn
- **Business Intelligence & Dashboarding:** Microsoft Power BI and Tableau for interactive metric visualization and reporting.

Conclusion

The Electric Vehicle Market & Performance Analysis project successfully maps out the technological state of modern electric mobility through empirical data from 478 vehicle models. SUVs dominate the market landscape with 244 models, far outpacing all other car body types. Technical evaluations reveal strong positive correlations between battery capacity and both driving range (0.8804) and fast-charging power (0.7286), confirming energy storage as the primary driver for extended range and higher charging rates. Furthermore, drivetrain analysis demonstrates distinct performance tiers, where All-Wheel Drive (AWD) leads across key metrics, averaging ~450 km driving range, ~210 km/h top speed, and ~4.6s acceleration, outperforming Rear-Wheel Drive (RWD) (~420 km, ~185 km/h, ~6.9s) and Front-Wheel Drive (FWD) (~300 km, ~155 km/h, ~9.7s).

Key performance benchmarks highlight individual category leaders: the Mercedes-Benz EQS 450+ achieves the highest overall driving range at 685 km, the Maserati GranTurismo Folgore sets the top speed benchmark at 325 km/h, and compact entries like the Dacia Spring Electric 45 (109 Wh/km) and Fiat 500e variants (112 Wh/km) define peak energy efficiency. Ultimately, this report demonstrates that while battery capacity dictates endurance and charging capability, optimizing energy efficiency alongside power density remains essential for advancing sustainable transportation across both compact and luxury performance segments.